MIL-C-37284A
28 August 1978
SUPERSEDING
MIL-C-37284
24 December 1975

MILITARY SPECIFICATION

CHLORPHENIRAMINE MALEATE TABLETS, MODIFIED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Chlorpheniramine Maleate Tablets, Modified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Shall be Chlorpheniramine Maleate Tablets, Modified, Timed, and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein:

S2. Classification. Shall be type XIII, class 2, style C, grade 4 (as specified below in S5.3).

S5.3 The following is added to this paragraph:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

Shall be coated tablets containing in each tablet, within the designated assay limits for the tablets, 8 mg of Chlorpheniramine Maleate, which shall be divided equally by a specially timed barrier to form an outer layer and an inner core. When the tablet is swallowed, the Chlorpheniramine Maleate in the outer layer shall be rapidly absorbed and the inner core shall release the medicament 4 to 6 hours later.

The inner cores shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required 4.0 mg of Chlorpheniramine Maleate, USP, per core, and the coated tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required 8.0 mg of Chlorpheniramine Maleate, USP, per coated tablet, when the cores and coated tablets are assayed as specified in 4.3.1.

S6.4.9.1 The coated tablets shall comply with the following disintegration test:

Using the USP Disintegration Test apparatus, test twenty (20) tablets in Simulated Gastric Fluid, USP. At the end of 1 hour, the color and the coating shall be completely removed and not more than one tablet shall indicate any evidence of pitting, disintegration, or dissolution of the enteric coated cores.

Replace the Simulated Gastric Fluid, USP, with Simulated Intestinal Fluid, prepared as specified below, and continue the test until all the tablets are disintegrated, recording time it takes for each tablet to disintegrate. The tablets shall disintegrate in the Simulated Intestinal Fluid in not less than 30 minutes and not more than 180 minutes, except that not more than two tablets may disintegrate in less than 30 minutes. In addition, the average disintegration time of all the tablets in the Simulated Intestinal Fluid shall be not less than 50 and not more than 120 minutes.

Disintegration time for the coated tablet shall be the point at which the enteric coating has broken and the inner core has dissolved or softened to the extent that any residue, other than the enteric coating shells, remaining on the screen of the tablet disintegrator basket, shall be a soft mass having no palpably firm part.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

Simulated Intestinal Fluid. Dissolve 15.0 grams of Sodium Bicarbonate in 250 ml of purified water and mix, add 2.8 grams of Pancreatin, mix well, and add sufficient purified water to make 1000 ml of fluid. Disks shall not be used in the disintegration test.

S6.4.11 The coated tablets shall contain not more than 4.0 percent of water when determined by the Karl Fischer Method as described in the USP, using sufficient accurately weighed, powdered tablets (approximately 5 grams).

3.1.1 Identification. Shall be determined as specified in 4.3.2.

3.1.2 Loss on drying. The loss on drying for the coated tablets shall be not more than 10.0 percent when accurately weighed, finely powdered tablets (approximately 5 grams) are dried at 105° C for 4 hours.

3.2 Ingredients.

3.2.1 Chlorpheniramine Maleate. The Chlorpheniramine Maleate used in the manufacture of the tablets shall be in accordance with the tests, standards and requirements of the USP, including any supplements or revisions thereto.

3.2.2 Other ingredients. All other ingredients shall comply with the requirements of S5.1 of FED-STD-140.

3.3 Expiration dating period. The expiration dating period shall be not less than 36 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with FED-STD-140.

4.3 Tests.

4.3.1 Assay.

Standard:

Weigh accurately 40 mg of USP Chlorpheniramine Maleate Reference Standard, transfer quantitatively into a 100 ml volumetric flask, dissolve and dilute to volume with purified water, and mix well: Reference Standard Solution.

Core:

Remove the coating from 20 whole tablets by washing them with purified water until the core is completely exposed (indicated by uniform, waxy appearance). Air-dry these cores, determine their average unit weight, then reduce them to a fine powder. Continue under "Coated" beginning with "Weigh accurately a portion. . . ."

Coated:

Determine the average unit weight of 20 whole tablets and reduce them to a fine powder. Weigh accurately a portion of the powder equivalent to 4 mg of chlorpheniramine maleate. Transfer the weighed portion into a suitable separator. Pipet 10.0 ml of Reference Standard Solution into a similar vessel and bring the volume of each to 80 ml with 0.1N hydrochloric acid. Shake mechanically for 10 minutes. Adjust the contents of each to pH 11 with 1N sodium hydroxide, cool to room temperature, add 100 ml of solvent hexane, and shake for 4 minutes. Let them stand until two distinct layers form, then transfer the aqueous phases (lower) into suitable separators and further extract each with a 25-ml portion of solvent hexane. Remove and discard the aqueous phases and combine the supernatant phases with their corresponding phases. Extract each with two 40-ml portions of 0.1N hydrochloric acid-shake each extraction manually for 3 minutes. Collect each extract (lower phase) in separate 100 ml volumetric flasks: 1 flask for reference standard, and the other for sample. Dilute each to volume with 0.1N hydrochloric acid and mix well.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

Obtain the absorption spectrum from 220 nm to 280 nm of each solution using a 1 cm silica cell and 0.1N hydrochloric acid as a blank on a suitable spectrophotometer. Record the values (Au for sample, As for reference standard) of maximum absorbance, at 264 ± 1 nm. Calculate the mg and percent of required amount of chlorpheniramine maleate in the core and in the coated tablet.

4.3.2 Identification test.

Finely powder a whole tablet and transfer a portion of the powder, equivalent to 4 mg of chlorpheniramine maleate, into a 250-ml separator. Add 50 ml of 0.1N hydrochloric acid, and shake for five minutes. Prepare a U.S.P. chlorpheniramine maleate reference standard solution containing 40 mg per 100 ml in 0.1N hydrochloric acid. Transfer 10 ml of this standard solution into a similar separator.

Render the contents of each separator alkaline (pH 11) with 1N sodium hydroxide and cool to room temperature. Add 100 ml of solvent hexane to each separator and shake the vessels for 2 minutes. Permit to stand until the layers separate. Withdraw and discard the lower (aqueous) phase. Add 100 ml of 0.1N hydrochloric acid to each vessel, shake for 3 minutes and permit to stand. After phase separation, withdraw and discard the first 10 ml to 15 ml of the lower layer from each vessel. Obtain the absorption spectrum (from 220 nm to 280 nm) of each remaining lower phase in a suitable spectrophotometer versus 0.1N hydrochloric acid, using 1 cm silica cells. The sample solution spectrum shall show a maximum and a minimum absorbance at the same wavelengths as that of the reference standard solution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

* 4.4 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

* 4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing one thousand (1000) tablets, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-B-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.9 of PPP-C-186.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twenty-four (24) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four (24) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

* 5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.4 Marking.

* 5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the national stock number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

* NOTES: (1) The national stock number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the national stock number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings shall be shown as specified in MIL-STD-129.

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking shall be shown as specified in MIL-STD-129.

* 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item :

National Stock No.	Item Identification
6505-00-655-8460	CHLORPHENIRAMINE MALEATE TABLETS, MODIFIED, Timed, 8 mg, 1000s

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6505-2050

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-37284A CHLORPHENIRAMINE MALEATE TABLETS, MODIFIED

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:21Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA. 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-21T10:21Z
Check the source to verify that this is the current version before use.